# Case Study: India Heatwave Intelligence Dashboard and Early Warning System (DSS)

Power BI project: four IMD stations, daily data, 1 Jan 1951 to 22 Jun 2024

## 1. Problem Statement & DSS Context

A single temperature reading says little about heat risk. 38°C is routine in Delhi in May, whereas Mumbai already treats 37°C as the heatwave threshold, and Bangalore has never recorded more than 38.9°C. The objective of this Business Intelligence (BI) initiative is to build a Decision Support System (DSS) that ingests 73 years of daily IMD station records, measures how far each day departs from the local normal, and classifies heat severity into Green, Yellow, Orange and Red tiers so that authorities can act before a heat event becomes a health emergency.

The DSS turns raw readings into decisions in three steps. A reading (Kolkata, 1 May 2024: 43.0°C) is compared with the 1991-2020 normal for that city and calendar day (34.6°C) to produce intelligence (a departure of +8.4°C), which the classification logic then converts into an operational decision (a Red alert, with hospital readiness, advisories and outdoor-work restrictions).

## 2. Data Preparation & ETL Architecture

The data foundation is four IMD city files (Delhi, Mumbai, Kolkata, Bangalore), each holding daily rainfall, maximum temperature and minimum temperature. Together they contain 107,346 city-day rows after the calendar is completed. All cleaning is done in Power Query through a reusable function (`fnCleanCity`) that is applied to every file in the data folder, so a new city file is picked up automatically on refresh.

- **Data Ingestion:** `Folder.Files` loads every `<city>-data.csv`, the city name is taken from the file name, and a `DataFolder` parameter keeps the source path configurable.
- **Mixed Date Formats:** most rows use dd-MM-yyyy text, but the last rows of the Delhi, Mumbai and Kolkata files are Excel serial numbers (for example 45463 = 19 Jun 2024). Both formats are converted to true dates.
- **Transmission Dropouts:** placeholders such as `---`, `----` and blanks become null. Trace rainfall (`Tr`, below 0.1 mm) is stored as 0 and flagged in a `RainTrace` column.
- **Sensor Anomalies:** physically impossible values are removed (Tmax outside 0 to 50°C, Tmin outside −5 to 40°C, Tmin at or above Tmax, rain above 500 mm in a day). This caught three readings, all dated 1 Dec 2023: Bangalore Tmin of 73.8°C, Kolkata rain of 1,014.5 mm and Mumbai rain of 1,011.7 mm.
- **Missing Days and Imputation:** the calendar is completed so that missing days appear as explicit "Missing" rows (January 2024 is absent for three cities). Gaps of 3 days or less are filled by forward-fill with the last valid reading and flagged "Imputed"; longer gaps stay empty rather than being invented.
- **Feature Engineering (The Normal Baseline):** for each city and calendar day, `NormalMaxTemp` is the 1991-2020 average of maximum temperature, computed only from original (unimputed) readings. The `Departure` column is the observed maximum minus this normal.

| Data quality flag | Rows | Share | Meaning |
|---|---|---|---|
| Original | 107,062 | 99.74% | Valid reading used as supplied |
| Imputed | 41 | 0.04% | Gap of 3 days or less, filled with last valid reading |
| Missing | 240 | 0.22% | No reading (80 days each for Delhi, Kolkata, Mumbai) |
| Outlier Removed | 3 | 0.00% | Impossible sensor value removed on 1 Dec 2023 |

**Table 2.1: Data quality profile after cleaning**

The semantic model links the weather fact table to a City dimension (state, coordinates, region and terrain) and a Date dimension (year, decade, month, IMD season, complete-year flag). Region and terrain are derived from latitude and longitude, and the terrain class decides which heatwave threshold applies.

### 3. Predictive Modeling & Classification in BI

The DSS uses deterministic, rule-based classification that follows IMD criteria. The threshold depends on terrain (Plains 40°C, Coastal 37°C, Hilly 30°C), and it is combined with the departure from the local normal.

| Alert | Status | Rule as implemented in the model |
|---|---|---|
| Green | Normal | Below the terrain threshold, or no heatwave condition met |
| Yellow | Hot day | Tmax at or above the threshold (40°C plains, 37°C coastal) but heatwave condition not met |
| Orange | Heatwave | Tmax at or above threshold and departure of 4.5°C or more, or Tmax of 45°C or more in plains |
| Red | Severe Heatwave | Tmax at or above threshold and departure above 6.4°C, or Tmax of 47°C or more in plains |

**Table 3.1: IMD-based alert tiers**

Of the 107,346 city-days, 96.2% are Green, 3.2% Yellow, 0.35% Orange (373 days), 0.04% Red (40 days) and 0.22% have no data. Four cities are mapped as Delhi (North Plains), Kolkata (East Plains), Bangalore (South Peninsula) and Mumbai (Coastal West, coastal threshold). The model also derives a 1 to 10 **Risk Score** from the peak temperature in the selection (10 at 47°C or more, 8 at 45°C, 6 at 40°C, 4 at 35°C) and a **Current Alert** for each station that looks at the last 30 days: Red if any severe day, Orange if any heatwave day, Yellow if any hot day, otherwise Green.

The cost of error in public heat risk is asymmetric. A false alarm costs a municipality some precautionary spending, while a missed heatwave means preventable heatstroke admissions and deaths. The thresholds are therefore deliberately sensitive: a coastal station alerts at 37°C, an absolute 45°C triggers an alert in the plains regardless of departure, and the 30-day Current Alert escalates on a single severe day. This prioritises early public safety over false-alarm avoidance.

### 4. Reporting, Authoring & Visual Analytics

The report has five main pages, a drill-through page and a hidden tooltip page. A year-range slicer on each main page lets authorities move between the full 1951 to 2024 record and a recent window, and a conditional-formatting palette (Green, Yellow, Orange, Red) is used consistently for alert states.

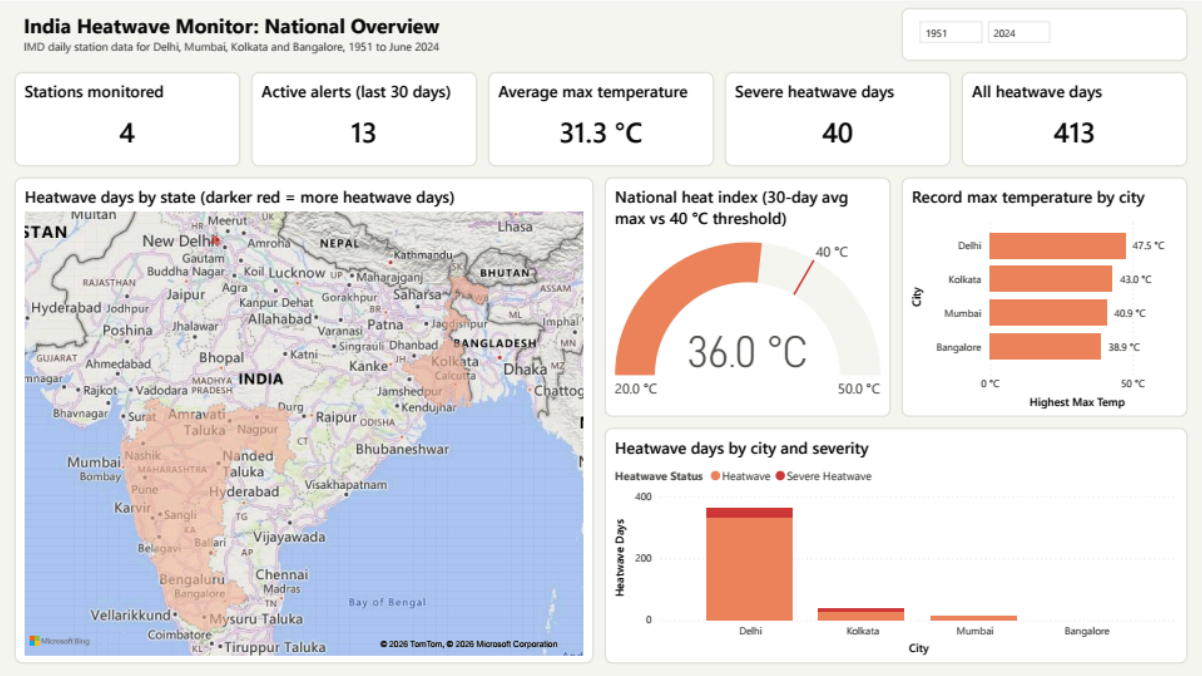



**Figure 4.1: National Overview, Heatwave Exposure and Record Extremes**

**Description:**

Page 1 combines five KPI cards (stations monitored, average maximum temperature, all heatwave days, severe heatwave days, active alerts in the last 30 days), a gauge of the national 30-day heat index against the 40°C plains threshold, a filled state map shaded by heatwave days, a column chart of heatwave days by city split by severity, and a bar chart of record maximum temperature by city.

**BI Decision Analysis:**

- The four stations logged 413 heatwave days, of which 40 were severe. Delhi alone accounts for 362 (88%) and 29 of the severe days, Kolkata has 37 (11 severe) and Mumbai 14. Bangalore has none.
- Record maxima show the same split: Delhi 47.5°C (9 Jun 1966), Kolkata 43.0°C (1 May 2024), Mumbai 40.9°C (28 Mar 2021) and Bangalore 38.9°C (24 Apr 2016). Kolkata's record is the newest, which points to emerging risk in the East Plains.
- The 30-day heat index across stations is 36.0°C as of 22 Jun 2024, below the 40°C gauge limit, yet all 13 active alerts sit in Delhi. The national average hides a local emergency.
- **Operational Impact:** serves as a Macro Resource Allocation view. Heat-health resources should be concentrated on Delhi and monitored in Kolkata rather than spread evenly across all stations.

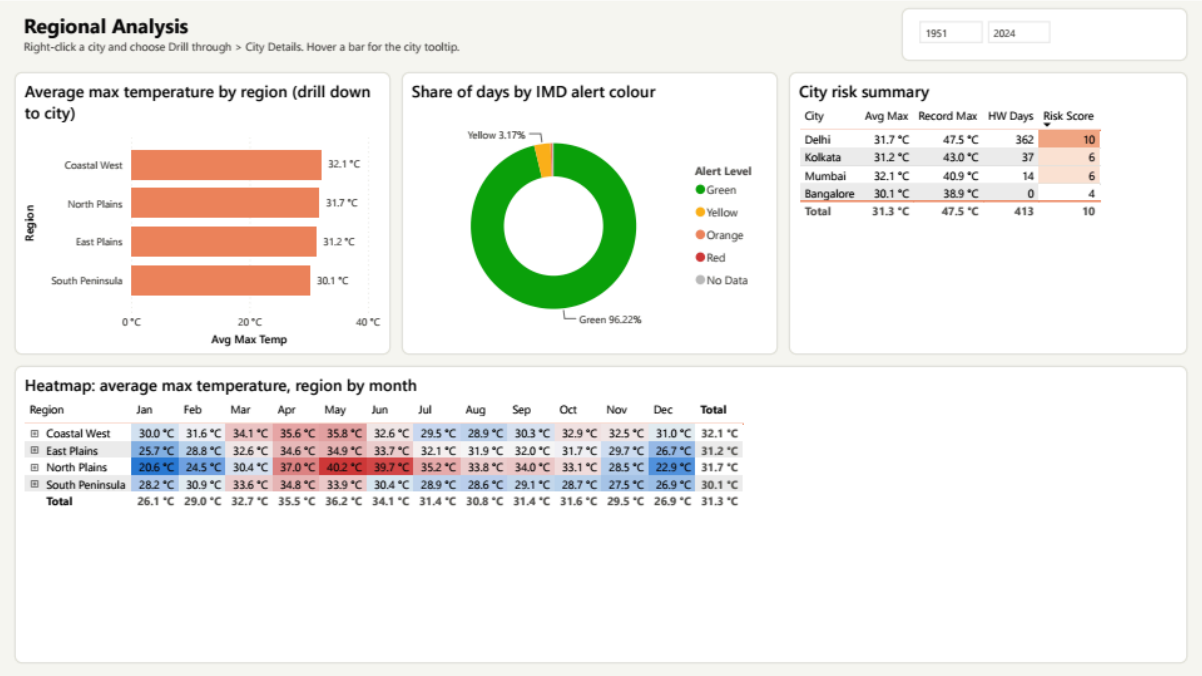



**Figure 4.2: Regional Analysis, Seasonal Profile and City Risk**

**Description:**

Page 2 holds a region-to-city drill-down bar chart of average maximum temperature, a region-by-month heatmap, a donut chart of the share of days by alert colour, and a city risk summary table (average maximum, record maximum, heatwave days and Risk Score). Hovering over the bar chart opens a custom tooltip page with average, record, heatwave days and risk score.

**BI Decision Analysis:**

- Average maximum temperature barely separates the stations: Mumbai 32.1°C, Delhi 31.7°C, Kolkata 31.2°C and Bangalore 30.1°C. Mumbai has the highest average but only 14 heatwave days, which shows why alerts must use departures from local normals instead of averages.
- The heatmap shows how different the thermal regimes are. Delhi swings from 20.6°C in January to 40.2°C in May (a range of 19.6°C), against 6.9°C for Mumbai, 7.9°C for Bangalore and 9.2°C for Kolkata.
- The donut shows that only 0.4% of all days reach Orange or Red, so a small number of days drive almost all of the risk. Delhi has 2,577 Yellow days and 362 Orange or Red days. Bangalore never leaves Green.
- The city risk scores are Delhi 10, Kolkata 6, Mumbai 6 and Bangalore 4.
- **Operational Impact:** region and month filters let planners schedule seasonal preparedness (Delhi from April, Kolkata April to June) instead of running year-round alert posture.

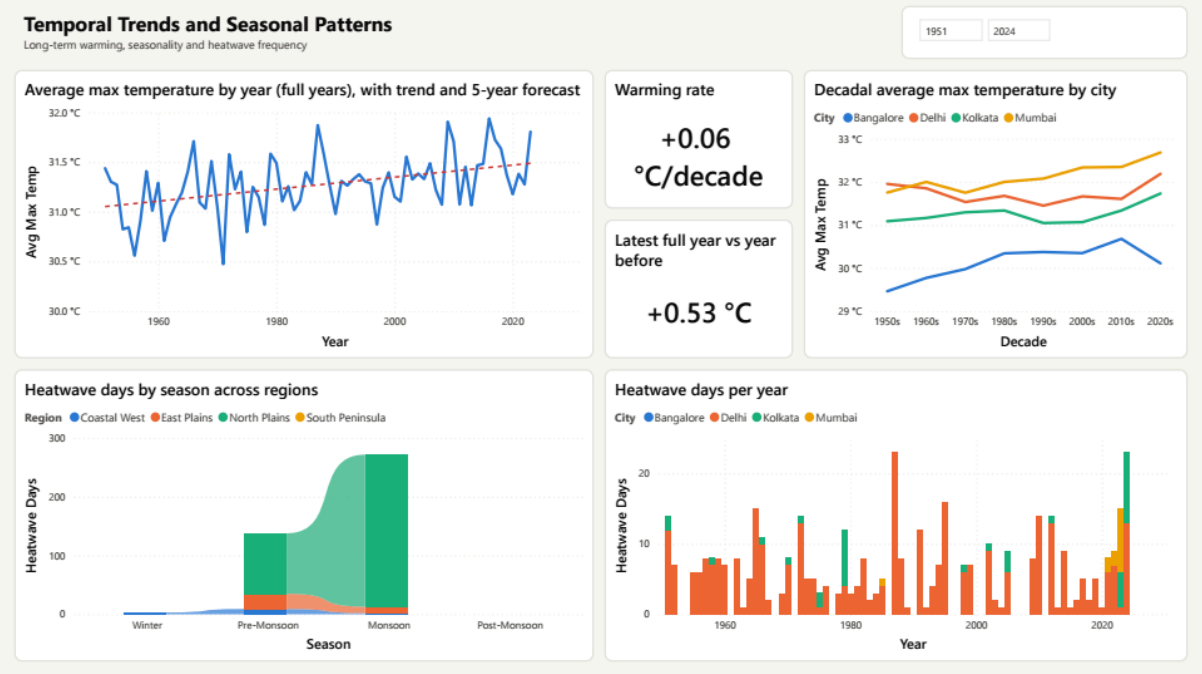



**Figure 4.3: Long-Term Trends, Decadal Change and Seasonality**

**Description:**

Page 3 tracks yearly heatwave days by city, decadal average maximum temperature by city, a ribbon chart of heatwave days by season and region, a warming-rate card, a latest-year change card, and a line chart of annual average maximum temperature with a linear trend and a 5-year forecast with a 95% band.

**BI Decision Analysis:**

- The four-station average maximum temperature is rising by about +0.06°C per decade across full years 1951 to 2023, and 2023 was +0.53°C warmer than 2022 (31.81°C against 31.28°C). The trend is uneven by city: Bangalore +0.14°C, Mumbai +0.11°C, Kolkata +0.03°C and Delhi −0.04°C per decade.
- Decadal averages for the 2020s are the highest on record for Delhi (32.19°C), Kolkata (31.73°C) and Mumbai (32.68°C), although the 2020s are only 4.5 years long and should be read with care.
- Heatwave counts are shifting geographically. Delhi has a steady 26 to 56 heatwave days per decade, but Kolkata has 15 in the 2020s against at most 12 in any earlier decade, and Mumbai has 13 in the 2020s against one in all earlier decades combined. The single worst years are 1987 and 2024 (23 days each; 2024 only runs to 22 June).
- Seasonally, 272 of 413 heatwave days fall in the IMD Monsoon months (190 in June alone) and 138 in Pre-Monsoon (March to May). June appears as Monsoon because seasons are assigned by calendar month, but Delhi's heat peaks before the monsoon arrives.
- **Operational Impact:** supports long-range planning (cooling-centre capacity, grid load, hospital staffing) and shows that heat preparedness must now extend to cities that historically had few alerts.

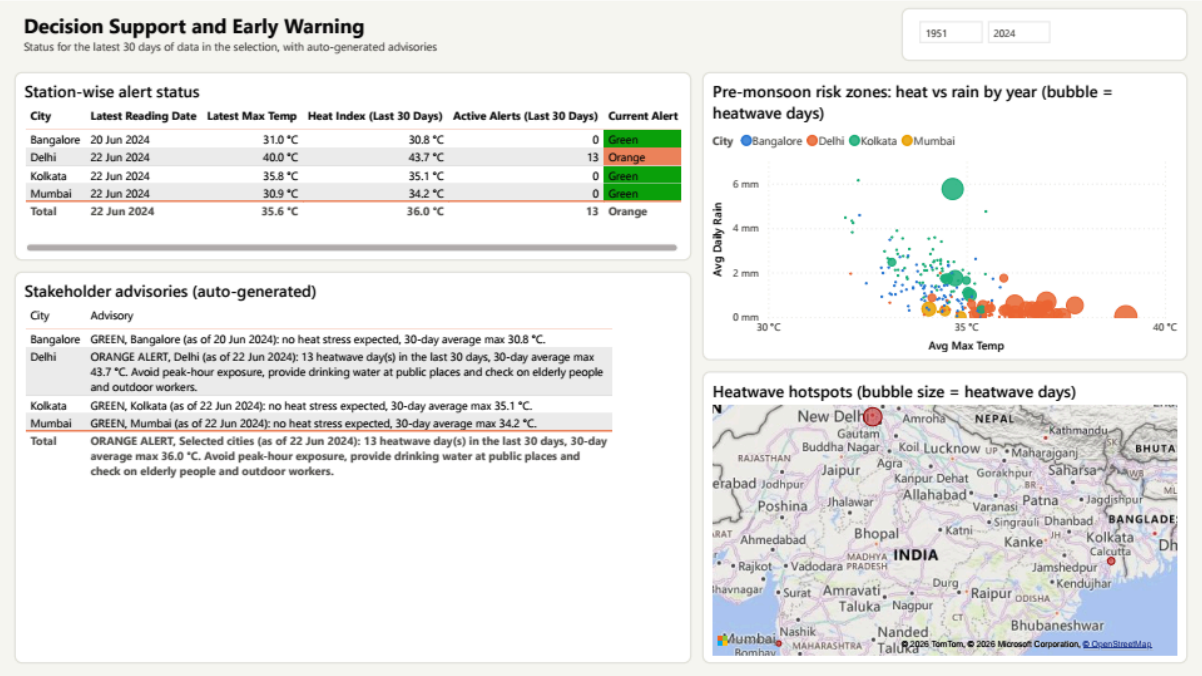



**Figure 4.4: Early Warning Operations Console**

**Description:**

Page 4 is the operational console. It shows a station-wise alert table (latest max, 30-day heat index, active alerts and colour-coded Current Alert), an auto-generated stakeholder advisory for each city, a bubble map of heatwave hotspots sized by heatwave days, and a scatter plot of average maximum temperature against average daily rainfall by year (bubble size equals heatwave days) to expose pre-monsoon risk zones.

**BI Decision Analysis:**

- As of 22 Jun 2024, Delhi is Orange: 13 heatwave days and 17 hot days in the last 30 days, a 30-day average maximum of 43.7°C and a latest reading of 40.0°C. Kolkata (35.1°C), Mumbai (34.2°C) and Bangalore (30.8°C) are Green. Bangalore's latest reading is dated 20 Jun 2024.
- The advisory text is generated by a DAX rule, so the wording escalates automatically with the alert tier and always carries the as-of date and the 30-day heat index, which removes manual drafting delay.
- The scatter shows heat and rainfall together: years with high temperatures and low rainfall are the pre-monsoon high-risk cases.
- **Operational Impact:** gives duty officers a single screen to decide which stations need advisories, hospital notifications or water-tanker rerouting today.

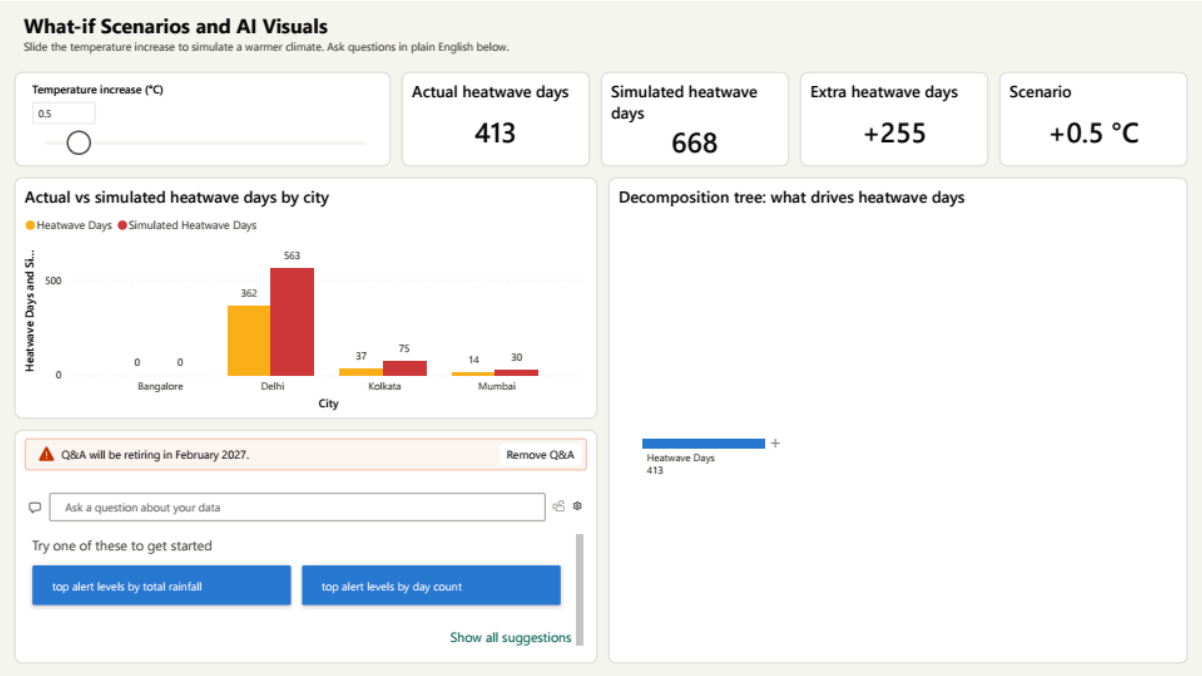



**Figure 4.5: What-if Warming Scenario and AI Visuals**

**Description:**
Page 5 contains a what-if slider (0 to 5°C in 0.5°C steps) that adds a uniform temperature increase to every observation and recomputes heatwave days under the same IMD rules. It shows actual, simulated and extra heatwave days in cards, an actual-versus-simulated column chart by city, a decomposition tree that breaks heatwave days down by city, decade, month, region and season, and a natural-language Q&A visual.

**BI Decision Analysis:**

| City | Actual | +1°C | +2°C | +3°C | +5°C |
|---|---|---|---|---|---|
| Delhi | 362 | 852 | 1,694 | 2,964 | 7,140 |
| Kolkata | 37 | 139 | 388 | 844 | 3,084 |
| Mumbai | 14 | 85 | 439 | 1,594 | 9,420 |
| Bangalore | 0 | 0 | 18 | 160 | 2,315 |
| **All stations** | **413** | **1,076** | **2,539** | **5,562** | **21,959** |

**Table 4.5: Simulated heatwave days under uniform warming**

- A uniform +1°C already raises heatwave days from 413 to 1,076 (2.6 times), and +2°C gives 2,539 (6.1 times). The response is strongly non-linear because many days sit just below the threshold.
- Coastal and southern stations are the most sensitive in relative terms. Mumbai goes from 14 to 439 days at +2°C, and Bangalore moves from zero to 18 days at +2°C and 160 days at +3°C.
- The decomposition tree confirms that heatwave exposure is concentrated: Delhi first, then the April to July window.
- **Operational Impact:** lets policy makers test infrastructure and staffing under warming scenarios and see which cities will first leave their historical risk class.

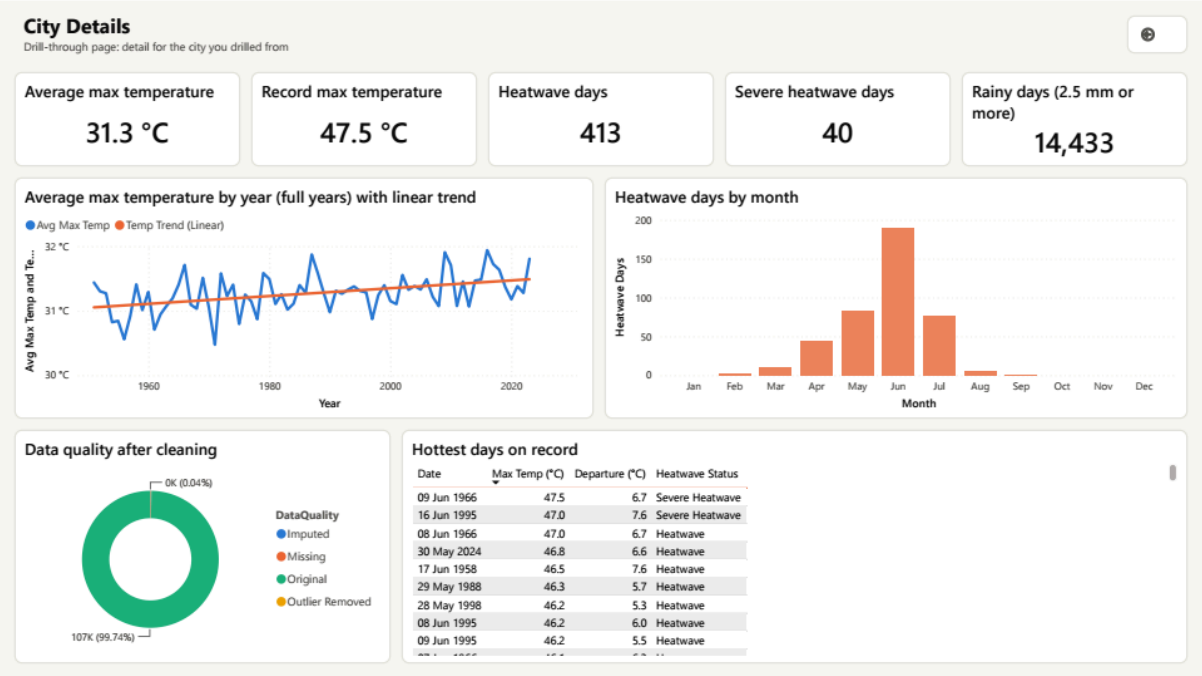



**Figure 4.6: City Details Drill-Through (Delhi example)**

**Description:**
Right-clicking a city on page 2 or page 4 and choosing Drill through opens the City Details page, filtered to that city. It shows cards for average maximum, record maximum, heatwave days, severe days and rainy days, a heatwave-days-by-month column chart, a yearly average maximum line with linear trend, a table of the hottest days on record, and a donut of data quality after cleaning.

**BI Decision Analysis:**

- For Delhi the page reports an average maximum of 31.7°C, a record of 47.5°C, 362 heatwave days (29 severe) and 2,630 rainy days (2.5 mm or more).
- Heatwave days by month peak in June (177), then July (76), May (71) and April (31), with only a handful in March, August and September. The top five hottest days are 9 Jun 1966 (47.5°C), 16 Jun 1995 (47.0°C), 8 Jun 1966 (47.0°C), 30 May 2024 (46.8°C) and 17 Jun 1958 (46.5°C). Severe days are concentrated in 1987 (9), 1982 (5) and 2010 (5).
- The quality donut keeps the data limits visible to the user: Delhi has 80 missing days and 7 imputed days.
- **Operational Impact:** lets a city officer inspect local history and timing before approving an advisory, instead of relying on the national view.

## 5. Operational Mitigation Strategy

The dashboard is not the final product. The final product is the action taken from the DSS output. The advisory measure maps each alert tier to a standard response:

- **Red (Severe Heatwave):** suspend outdoor work between 12:00 PM and 4:00 PM, open cooling centres, put hospitals on heatstroke readiness and issue public SMS warnings.
- **Orange (Heatwave):** avoid peak-hour exposure, provide drinking water at public places and check on elderly people and outdoor workers.
- **Yellow (Hot Day):** stay hydrated and follow IMD bulletins.
- **Green:** no heat stress expected; continue monitoring.

Access is controlled with row-level security. A **North Region Analyst** role sees only the North Plains stations and a **Coastal Region Analyst** role sees only coastal stations, so regional teams work from the data they are responsible for.

## 6. Data Limitations and Assumptions

- Only four stations are covered, so the results should not be generalised to all of India.
- Departures use a single 1991-2020 normal for every year. Readings from the 1950s are therefore judged against a modern baseline.
- Mumbai rainfall exists only from 2021 onward (1,183 readings), so rainfall visuals for Mumbai are sparse.
- January 2024 is missing for Delhi, Kolkata and Mumbai, and 2024 ends on 22 June, so 2024 totals are partial.
- In plains, a day of 45.0 to 46.9°C is classed Heatwave even when its departure exceeds 6.4°C; this affects 12 Delhi days that would otherwise be Severe.

## 7. Conclusion

The India Heatwave Intelligence DSS moves heat management from reactive to proactive. By combining a cleaned 73-year record, local 1991-2020 normals, terrain-specific IMD thresholds and an automated advisory engine, it flags risk that national averages hide: Delhi carries 88% of heatwave days today, Kolkata and Mumbai are showing recent increases, and a modest 1 to 2°C of warming would multiply heatwave days several times over. The dashboard gives authorities one place to see where risk is concentrated, when it peaks and what action to take.